

SandCat Overhead Controller — Point-to-Point Field Wiring (bit-level)

Every terminal & wire, in conductor color. CONTROLLER (OPTO ch · 3724 port.bit) ◀ TB ▶ FIELD. White OPTO=inputs (50-pin ribbon), Red=outputs. Source: DCH Field Wiring + firmware.
Map: OPTO IN ch N = 3724 bit N (A0=0-7, B0=8-15, C0=16-23); count=even ch, zero=odd ch. OPTO OUT ch N → A1/B1/C1, DROP-(N+1).

FUSE	CONTROLLER (OPTO ch · 3724 port.bit · var)	TB	WIRE → FIELD DEVICE / SIGNAL
SCALE 1 — sync signals + RS-422 load cell (firmware sync #0)			
F-1 3A	PULS +24V		RED +24 VDC to Scale-1
	−24V common	TB-24	BLACK −24 VDC to Scale-1
	IN ch0 · 3724 A0.0 · sync_in	TB-1	WHITE SD signal Scale-1 (shackle / count)
	IN ch1 · 3724 A0.1 · sync_zero	TB-2	GREEN ZD signal Scale-1 (zero)
	DB-9 p3 → PC COM (RS-422)	TB-3	GREEN Load cell Scale-1 TXD−
	DB-9 p8 → PC COM (RS-422)	TB-4	BLACK Load cell Scale-1 RXD+
	DB-9 p7 → PC COM (RS-422)	TB-5	WHITE Load cell Scale-1 TXD+
	DB-9 p2 → PC COM (RS-422)	TB-6	RED Load cell Scale-1 RXD−
	DB-9 GND	TB-SHLD	BARE Load cell Scale-1 SHIELD
SCALE 2 — sync signals + RS-422 load cell (firmware sync #1)			
F-2 3A	PULS +24V		RED +24 VDC to Scale-2
	−24V common	TB-24	BLACK −24 VDC to Scale-2
	IN ch2 · 3724 A0.2 · sync_in	TB-7	WHITE SD signal Scale-2 (shackle / count)
	IN ch3 · 3724 A0.3 · sync_zero	TB-8	GREEN ZD signal Scale-2 (zero)
	DB-9 p3 → PC COM (RS-422)	TB-9	GREEN Load cell Scale-2 TXD−
	DB-9 p8 → PC COM (RS-422)	TB-10	BLACK Load cell Scale-2 RXD+
	DB-9 p7 → PC COM (RS-422)	TB-11	WHITE Load cell Scale-2 TXD+
	DB-9 p2 → PC COM (RS-422)	TB-12	RED Load cell Scale-2 RXD−
	DB-9 GND	TB-SHLD	BARE Load cell Scale-2 SHIELD
DROP SYNCs 1–6 (firmware sync #2–#7) — each: fused +24, common −24, SD + ZD			
F-3 3A	PULS +24V		RED +24 VDC to Drop Sync-1
	−24V common	TB-24	BLACK −24 VDC to Drop Sync-1
	IN ch4 · 3724 A0.4 · sync_in	TB-13	WHITE SD signal Drop Sync-1 (shackle)
	IN ch5 · 3724 A0.5 · sync_zero	TB-14	GREEN ZD signal Drop Sync-1 (zero)
F-4 3A	PULS +24V		RED +24 VDC to Drop Sync-2
	−24V common	TB-24	BLACK −24 VDC to Drop Sync-2
	IN ch6 · 3724 A0.6 · sync_in	TB-15	WHITE SD signal Drop Sync-2 (shackle)
	IN ch7 · 3724 A0.7 · sync_zero	TB-16	GREEN ZD signal Drop Sync-2 (zero)
F-5 3A	PULS +24V		RED +24 VDC to Drop Sync-3
	−24V common	TB-24	BLACK −24 VDC to Drop Sync-3
	IN ch8 · 3724 B0.0 · sync_in	TB-17	WHITE SD signal Drop Sync-3 (shackle)
	IN ch9 · 3724 B0.1 · sync_zero	TB-18	GREEN ZD signal Drop Sync-3 (zero)
F-6 3A	PULS +24V		RED +24 VDC to Drop Sync-4
	−24V common	TB-24	BLACK −24 VDC to Drop Sync-4
	IN ch10 · 3724 B0.2 · sync_in	TB-19	WHITE SD signal Drop Sync-4 (shackle)
	IN ch11 · 3724 B0.3 · sync_zero	TB-20	GREEN ZD signal Drop Sync-4 (zero)

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FUSE	CONTROLLER (OPTO ch · 3724 port.bit · var)	TB	WIRE → FIELD DEVICE / SIGNAL
F-7 3A	PULS +24V		RED +24 VDC to Drop Sync-5
	−24V common	TB-24	BLACK −24 VDC to Drop Sync-5
	IN ch12 · 3724 B0.4 · sync_in	TB-21	WHITE SD signal Drop Sync-5 (shackle)
	IN ch13 · 3724 B0.5 · sync_zero	TB-22	GREEN ZD signal Drop Sync-5 (zero)
F-8 3A	PULS +24V		RED +24 VDC to Drop Sync-6
	−24V common	TB-24	BLACK −24 VDC to Drop Sync-6
	IN ch14 · 3724 B0.6 · sync_in	TB-23	WHITE SD signal Drop Sync-6 (shackle)
	IN ch15 · 3724 B0.7 · sync_zero	TB-24	GREEN ZD signal Drop Sync-6 (zero)
GRADE — grade sync + 3 grade detects (3724 Port C0)			
F-9 3A	PULS +24V		RED +24 VDC to Grade Sync
	−24V common	TB-24	BLACK −24 VDC to Grade Sync
	IN ch16 · 3724 C0.0 · switch_in	TB-25	WHITE SD signal Grade Sync (shackle)
	IN ch17 · 3724 C0.1 · switch_in	TB-26	GREEN ZD signal Grade Sync (zero)
F-10 3A	PULS +24V		RED +24 VDC to Grade Detects
	−24V common	TB-24	BLACK −24 VDC to Grade Detects
	IN ch18 · 3724 C0.2 · switch_in	TB-27	GRAY Grade Detect-1
	IN ch19 · 3724 C0.3 · switch_in	TB-28	GRAY Grade Detect-2
	IN ch20 · 3724 C0.4 · switch_in	TB-29	GRAY Grade Detect-3
	IN ch21 · 3724 C0.5 · switch_in	OPTO 21	GRAY POWER-LOSS / SHUTDOWN — external-power-lost signal from the DC-UPS → graceful shutdown
	(spare)	TB-30	GRAY NO CONNECTION
MISSED-BIRD DETECTS (3724 C0.6 / C0.7 = MB_BIT1 / MB_BIT2) — note: C0.5 = power-loss input (no 2nd grade sync on this build)			
F-11	PULS +24V		RED +24 VDC to Missed-Bird Detects
	−24V common	TB-24	BLACK −24 VDC to Missed-Bird Detects
	IN ch22 · 3724 C0.6 · switch_in	TB-31	GRAY Missed-Bird Detect-1
	IN ch23 · 3724 C0.7 · switch_in	TB-32	GRAY Missed-Bird Detect-2
DROP (KICKER) OUTPUTS 1–24 (RED OPTO ch N → 3724 A1/B1/C1, active-low → solenoid)			
F-12 5A	PULS +24V		RED +24 VDC to Drops 1-8
	OUT ch0 · 3724 A1.0 · output_byte	TB-33	BLACK DROP-1 kicker solenoid
	OUT ch1 · 3724 A1.1 · output_byte	TB-34	BLACK DROP-2 kicker solenoid
	OUT ch2 · 3724 A1.2 · output_byte	TB-35	BLACK DROP-3 kicker solenoid
	OUT ch3 · 3724 A1.3 · output_byte	TB-36	BLACK DROP-4 kicker solenoid
	OUT ch4 · 3724 A1.4 · output_byte	TB-37	BLACK DROP-5 kicker solenoid
	OUT ch5 · 3724 A1.5 · output_byte	TB-38	BLACK DROP-6 kicker solenoid
	OUT ch6 · 3724 A1.6 · output_byte	TB-39	BLACK DROP-7 kicker solenoid
	OUT ch7 · 3724 A1.7 · output_byte	TB-40	BLACK DROP-8 kicker solenoid
F-13 5A	PULS +24V		RED +24 VDC to Drops 9-16
	OUT ch8 · 3724 B1.0 · output_byte	TB-41	BLACK DROP-9 kicker solenoid

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FUSE	CONTROLLER (OPTO ch · 3724 port.bit · var)	TB	WIRE → FIELD DEVICE / SIGNAL
	OUT ch9 · 3724 B1.1 · output_byte	TB-42	BLACK DROP-10 kicker solenoid
	OUT ch10 · 3724 B1.2 · output_byte	TB-43	BLACK DROP-11 kicker solenoid
	OUT ch11 · 3724 B1.3 · output_byte	TB-44	BLACK DROP-12 kicker solenoid
	OUT ch12 · 3724 B1.4 · output_byte	TB-45	BLACK DROP-13 kicker solenoid
	OUT ch13 · 3724 B1.5 · output_byte	TB-46	BLACK DROP-14 kicker solenoid
	OUT ch14 · 3724 B1.6 · output_byte	TB-47	BLACK DROP-15 kicker solenoid
	OUT ch15 · 3724 B1.7 · output_byte	TB-48	BLACK DROP-16 kicker solenoid
F-14 5A	PULS +24V		RED +24 VDC to Drops 17-24
	OUT ch16 · 3724 C1.0 · output_byte	TB-49	BLACK DROP-17 kicker solenoid
	OUT ch17 · 3724 C1.1 · output_byte	TB-50	BLACK DROP-18 kicker solenoid
	OUT ch18 · 3724 C1.2 · output_byte	TB-51	BLACK DROP-19 kicker solenoid
	OUT ch19 · 3724 C1.3 · output_byte	TB-52	BLACK DROP-20 kicker solenoid
	OUT ch20 · 3724 C1.4 · output_byte	TB-53	BLACK DROP-21 kicker solenoid
	OUT ch21 · 3724 C1.5 · output_byte	TB-54	BLACK DROP-22 kicker solenoid
	OUT ch22 · 3724 C1.6 · output_byte	TB-55	BLACK DROP-23 kicker solenoid
	OUT ch23 · 3724 C1.7 · output_byte	TB-56	BLACK DROP-24 kicker solenoid
F-15 5A	PULS +24V		RED +24 VDC spare (reserved)
INCOMING POWER (TB-1 end)			
F-16 6A	Main fuse		Incoming power — MAIN FUSE
		TB-GND	BARE Incoming power — GROUND
		TB-L2	WHITE Incoming power — 120 VAC NEUTRAL
		TB-L1	BLACK Incoming power — 120 VAC HOT
TB-2 — LOGIC POWER (internal distribution, no field landing)			
F-17 3A		TB+5	RED +5 VDC to CPU & I/O logic
		TB-5	BLACK −5 VDC to CPU & I/O logic
F-18 3A			RED +24 VDC to OPTO boards
F-19 3A			RED +24 VDC to monitor
	24V buss	TB+24 ×5	RED 24 VDC positive buss (no field)
	24V common	TB−24 ×5	BLACK 24 VDC common buss (no field)
		TB-GND ×2	BARE Ground